

Base Models Look Human To AI Detectors

Yixuan (Even) Xu

Ziqian Zhong

Aditi Raghunathan

Fei Fang

J. Zico Kolter

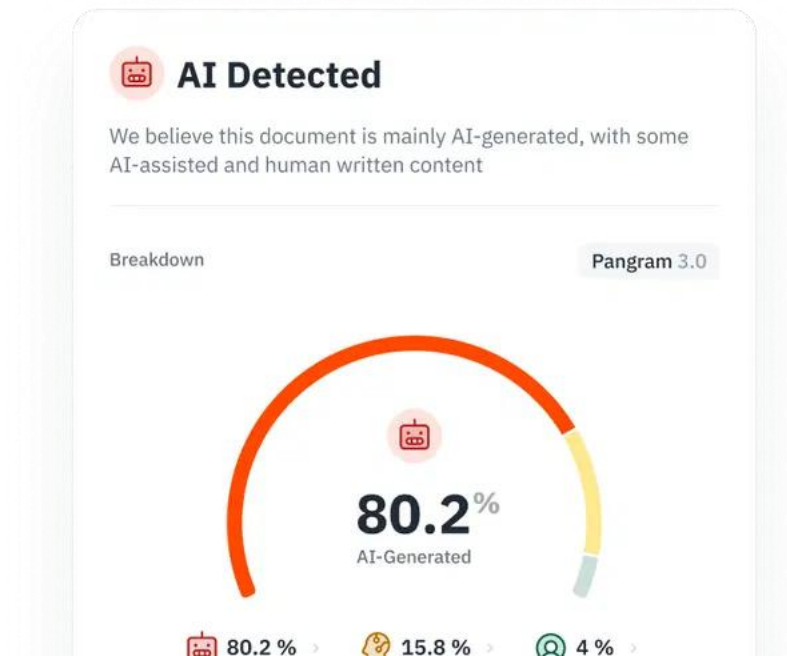
Carnegie Mellon University

AI Text Detection

- AI-generated text is entering the real-world at scale
- In education and academic-integrity workflows, one might want to distinguish between human-written and AI-generated text
- **AI text detection:** Use *patterns* in generated text to do this
 - *“Certainly! Here is”*
 - *Em-dashes, semicolons*
 - *Other statistical patterns*

Commercial Detectors

- Great interest in practice, already turned into products
- GPTZero and Pangram (show in the right) are two of them
 - *Many more: Quillbot, Scribbr, ZeroGPT, Writer, Copyleaks, Undetectable AI, Originality.ai, Sapling.ai, Turnitin*
- In use in some CMU courses



Detector Evasion

- A growing body of academic work
 - Adversarial paraphrasing
 - RL with detector feedback
 - Other detector-aware optimizations
- Many of the commercial detector providers also has “*humanizers*”
 - *To make model outputs not recognized by their own detectors, often generalizes*

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The Motivating Question

*Are there **models** whose outputs commercial detectors already **judge to be human-written**, **without detector-aware optimization**?*

A Surprisingly Simple Answer

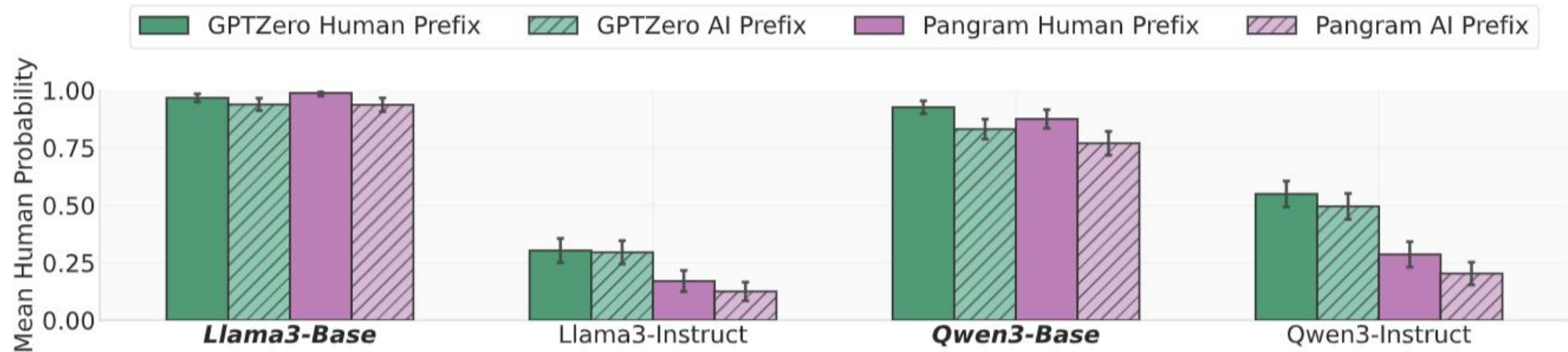
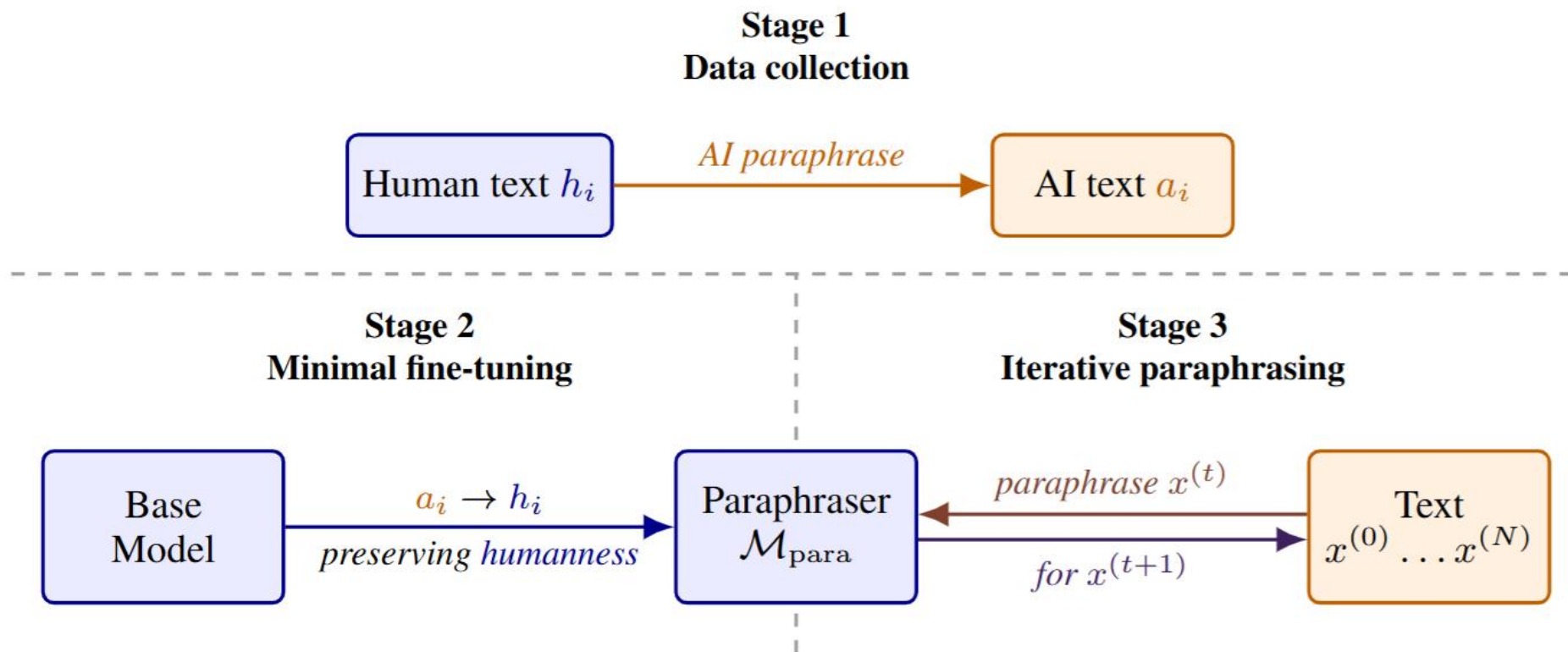


Figure 1: GPTZero and Pangram human-probability scores on text generated by base and instruction-tuned Llama3-8B and Qwen3-8B models when conditioned on human-written or AI-generated prefixes. The error bars show 95% confidence intervals. Across both model families, base-model continuations are judged substantially more human than instruct-model continuations.

What Are We Seeing Here?

- These commercial detectors assign **significantly higher** human probabilities to base models than to instruct models
 - *Key thesis: “Base models look human to AI detectors”*
 - *So yes to the previous questions, base models simply do satisfy it*
- Human-written prefixes make model continuations look **slightly more human** than AI prefixes
- Can we somehow **use** these observations?

Humanization by Iterative Paraphrasing



Data Collection

- We would like to construct pairs of texts (a_i, h_i) , such that a_i is **AI** text, h_i is **human text**, and their **meanings are similar**
- To do this, we
 - Collect **pieces of human-written texts** h_i
 - Let AI **paraphrase them** to obtain the **AI counterparts** a_i
- In the experiments, we use **MAGE** and **RAID**, as the **human text** source, and uses **GPT-5-nano** as the **paraphraser**

Minimal Fine-Tuning

- We want to make base models able to paraphrase AI texts, while causing less distortion
 - *To preserve its humanness*
- We do this with a minimal fine-tuning format (right)
 - Condition on the AI text
 - Loss only on the human text

```
<source_text>  
    AI paraphrase a  
</source_text>  
  
<target_text>  
    Original human passage h  
</target_text>
```

Iterative Paraphrasing

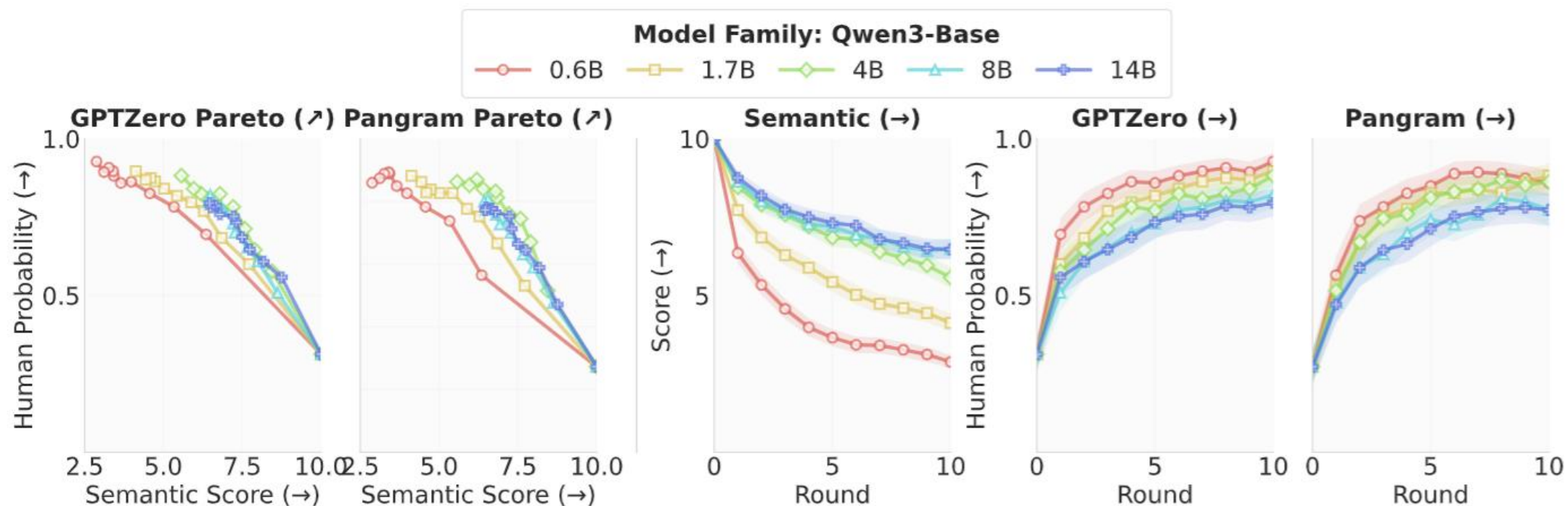
- We have previously observed that conditioning on a human prefix makes model output slightly more human
- Iterative paraphrasing
 - Use the paraphraser multiple times
 - Gradually “*wash away*” the AI-ness

```
for  $t \in \{1, \dots, N\}$  do  
     $p^{(t)} \leftarrow \text{FORMATASPROMPT}(x^{(t-1)})$   
     $x^{(t)} \leftarrow \mathcal{M}_{\text{para}}(p^{(t)})$   
return  $x^{(N)}$ 
```

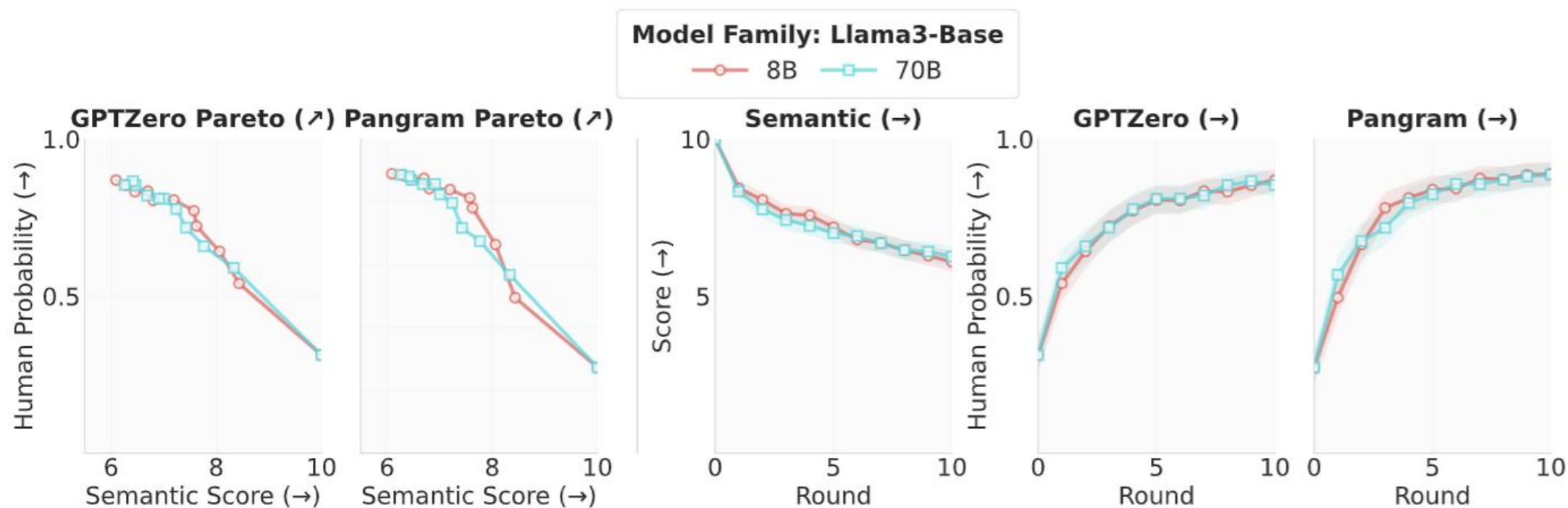
Experiments

- Datasets
 - **Sources:** MAGE and RAID, both contain **human** and **AI** texts
 - **Training set:** $\{(a_i, h_i)\}$, h_i from the sources, a_i their paraphrases
 - **Evaluation set:** Collection of **AI** texts from the sources
- Evaluation metrics
 - **Semantic scores:** Semantic closeness judged by GPT-5-nano
 - **Detector scores:** Human probability judged by Pangram or GPTZero

Model Family: Qwen3-Base



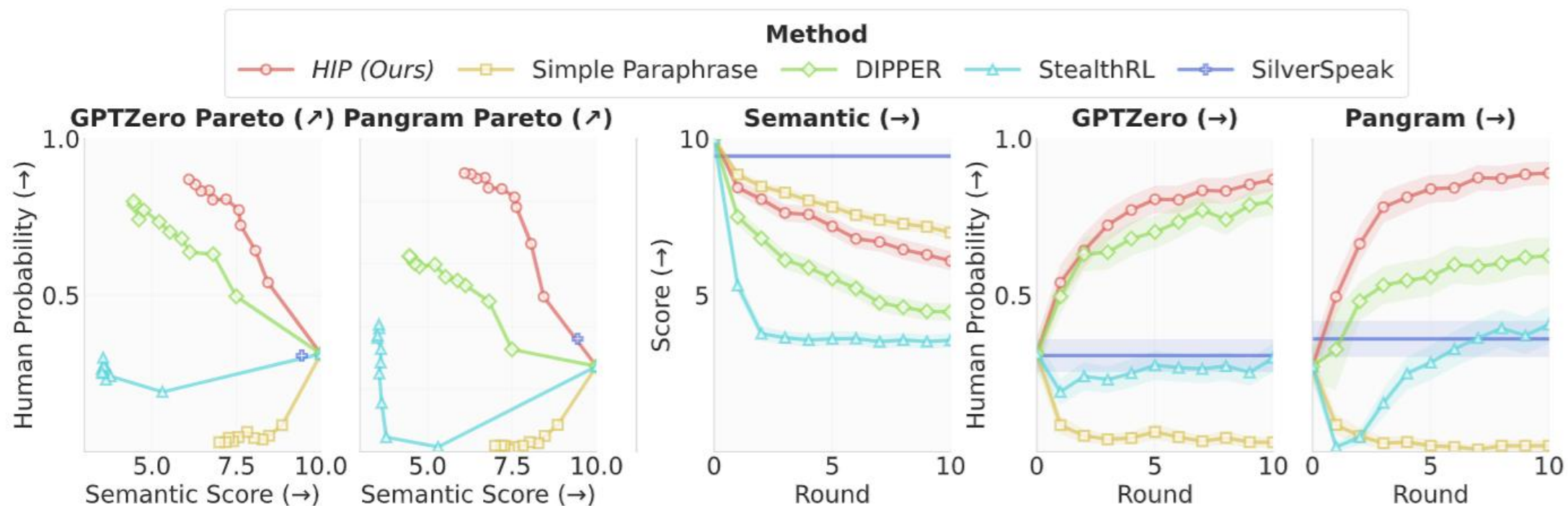
Model Family: Llama3-Base



Comparing With Baselines

- We compare HIP with representative detector-evasion baselines
 - **Simple Paraphrase:** Zero-shot paraphrasing at inference time
 - **DIPPER:** Supervised paraphrasing method that aims to preserve meaning while varying surface form
 - **SilverSpeak:** Perturbing token appearance without rewriting the text, targeting detector sensitivity to character-level cues
 - **StealthRL:** RL-based detector evasion method that optimizes a paraphraser against open-source detectors

Comparing With Baselines



Example: Original Text

Semantic: 10 (ref.)

GPTZero: 0.0%

Pangram: 0.0%

A police force has apologised after its officers made a sandcastle “crime scene” featuring a naked dead woman. The incident happened during a training exercise on Bournemouth beach in Dorset, England. The force said the scene was “clearly inappropriate” and an investigation has been launched. It is not yet clear how the scene came to be created or who was responsible. The incident has caused outrage on social media, with many people calling for the officers involved to be disciplined. This is not the first time that the police have been accused of being insensitive in their training exercises. In 2016, the Metropolitan Police in London was criticised for using photos of real victims of knife crime in a training session.

Example: HIP Round 1

Semantic: 8

GPTZero: 8.2%

Pangram: 0.0%

A police force has apologised after its officers created a “crime scene” sandcastle, including a naked dead woman. Dorset Police said officers from its rural crime unit were taking part in a training exercise on Bournemouth beach in the south west of England. The force said the scene was “clearly inappropriate” and an investigation was under way. It is not yet known how the sandcastle was created or who was responsible. The incident has been condemned on social media and people have called for the officers involved to be disciplined. It is not the first time police have been accused of being insensitive in their training. In 2016, the Metropolitan Police in London faced criticism for using photos of actual victims of knife crime in a training exercise.

Example: HIP Round 2

Semantic: 9

GPTZero: 20.9%

Pangram: 100.0%

A police force has apologised for the actions of officers who created a “crime scene” sandcastle including a naked dead woman. Dorset Police said officers from its rural crime unit were taking part in a training exercise on Bournemouth beach in the south west of England. The force said it was an “inappropriate” scene and was investigating. It is not known how the sandcastle was made and who was responsible. The incident has been criticised on social media, with people calling for the officers involved to be disciplined. It is not the first time police have been criticised for appearing insensitive in their training. In 2016, the Metropolitan Police in London was criticised for using photographs of victims of knife crime in their training.

Example: HIP Round 10

Semantic: 9

GPTZero: 100.0%

Pangram: 100.0%

Dorset Police has apologised after officers were filmed building a sandcastle. It included the words crime scene and a naked dead woman. Dorset Police said it was “appalled” officers were caught on camera building the sandcastle on a rural crime team training day. The officers were at Bournemouth beach, in the south of England. Dorset Police said it was investigating the “inappropriate scene.” It was not clear how the sandcastle was constructed or how many officers were involved. Officers were criticised on social media, with many saying those involved should be disciplined. Police have been accused of being insensitive in the way they treat victims of crime and their families. The Met police in London was criticised heavily in November 2018 for using images of victims of knife crime in their training.

Implications

- “Base Models Look Human To AI Detectors”
 - Current detectors are not primarily detecting “AI” in general
 - They appear to be sensitive to the statistical effects of post-training
- Questions for future research
 - Does post-training inherently leave detector-visible non-human statistical fingerprints? Can we avoid them?
 - Are base model outputs really indistinguishable from human texts?

Our Contributions



- We introduce a simple but surprising observation: *Base model outputs look human to state-of-the-art AI detectors*
- Building on this observation, we propose HIP, an intervention that adapts a base model into an AI text paraphraser
 - Empirically, HIP works across model families and sizes
 - And yields a stronger semantic-evasion trade-off than the prior work